

AI-POWERED NETWORK INTRUSION DETECTION SYSTEM (NIDS)

NETWORK INTRUSION DETECTION SYSTEM USING MACHINE LEARNING

MILESTONE -2(WEEK-3 AND WEEK-4)

NAME :VENNELA RAGHAVA KONDURU



PROJECT OVERVIEW

- Goal: Build an AI-based **Network Intrusion Detection System**
- Dataset: **NSL-KDD / CICIDS2017**
- Detect multiple attack classes
 - Normal
 - DoS
 - Probe
 - R2L
 - U2R

FEATURE ENGINEERING WORKFLOW

- Categorical Encoding
- Feature Matrix Preparation
- Noise Removal
- Feature Importance Ranking
- Feature Scaling
- Class Imbalance Handling
- Dimensionality Reduction

CATEGORICAL ENCODING

- Encoding techniques used:
- **One-Hot Encoding**
- Applied to protocol_type and flag
- Creates binary columns for each category
- **Frequency Encoding**
- Applied to service feature
- Replaces values with occurrence frequency
- **Important Rule**
- Encoder fitted on **train data only**

FEATURE MATRIX PREPARATION

- Steps performed:
- Target column `attack_class` encoded using **LabelEncoder**
- Removed metadata columns:
 - `label`
 - `difficulty_level`
 - `attack_class`
 - `is_outlier`
- Created feature matrices:
- Test aligned with train columns using **reindex**

NOISE REMOVAL

- High correlation causes redundancy.
- Method used:
- Computed **correlation matrix**
- Removed features with **correlation > 0.95**
- Why?
- Highly correlated features carry duplicate information
- Removing them reduces noise and improves model efficiency

FEATURE SCALING

- Scaling method used:
- **RobustScaler**
- Why RobustScaler?
- Uses **median and IQR**
- Less sensitive to outliers
- Better for network traffic data
- Important rule:
- Fit on **train**
- Apply to **test**

HANDLING CLASS IMBALANCE (SMOTE)

- Problem:
- DoS attacks → many samples
- U2R attacks → very few samples
- Solution: SMOTE (Synthetic Minority Oversampling Technique)
- What it does:
- Generates synthetic samples
- Balances minority classes
- Applied only to: **Training dataset**

PCA

- PCA helps visually verify whether attack classes are separable in the feature space.
- Used **Principal Component Analysis**
- Purpose:
- Visualize class separation
- Reduce dimensionality

SUPERVISED MODEL TRAINING

- Models trained:
- **Random Forest**
- **Support Vector Machine (SVM)**
- **Logistic Regression**
- Evaluation Metrics:
- Accuracy
- Precision
- Recall
- F1 Score
- These models learn patterns in network traffic and classify whether the activity is normal or malicious.

CONCLUSION

- ❑ Successfully performed feature engineering and feature selection on the network intrusion dataset.
- ❑ Removed noisy, redundant, and low-variance features to improve data quality.
- ❑ Applied Random Forest feature importance to identify the most relevant features.
- ❑ Handled class imbalance using SMOTE to improve detection of rare attacks.
- ❑ Used PCA visualization to understand class separation.
- ❑ Prepared a balanced and optimized dataset for supervised machine learning models.
- ❑ Trained models such as Random Forest, SVM, and Logistic Regression and
- ❑ evaluated them using Accuracy, Precision, Recall, and F1-Score.